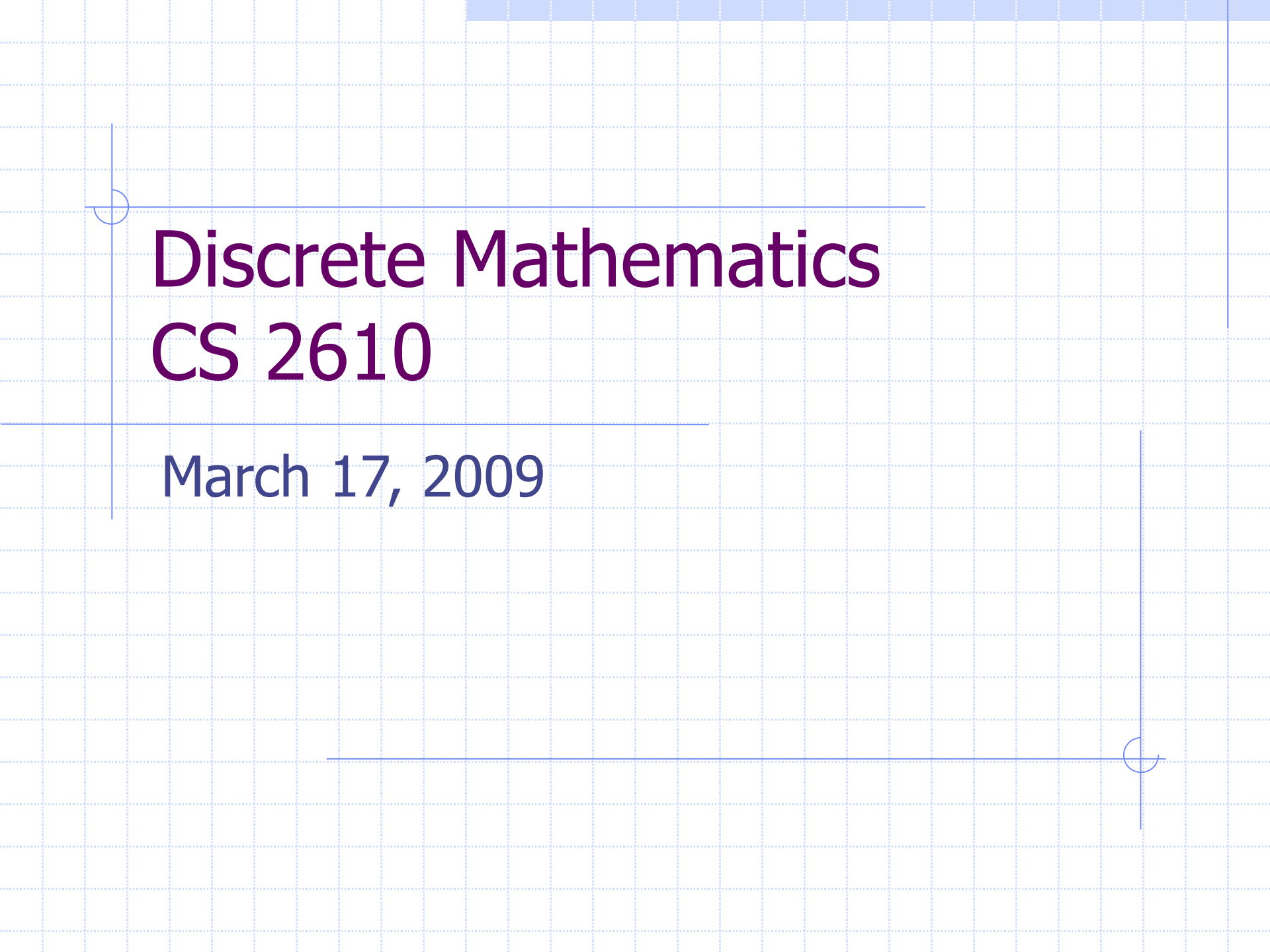
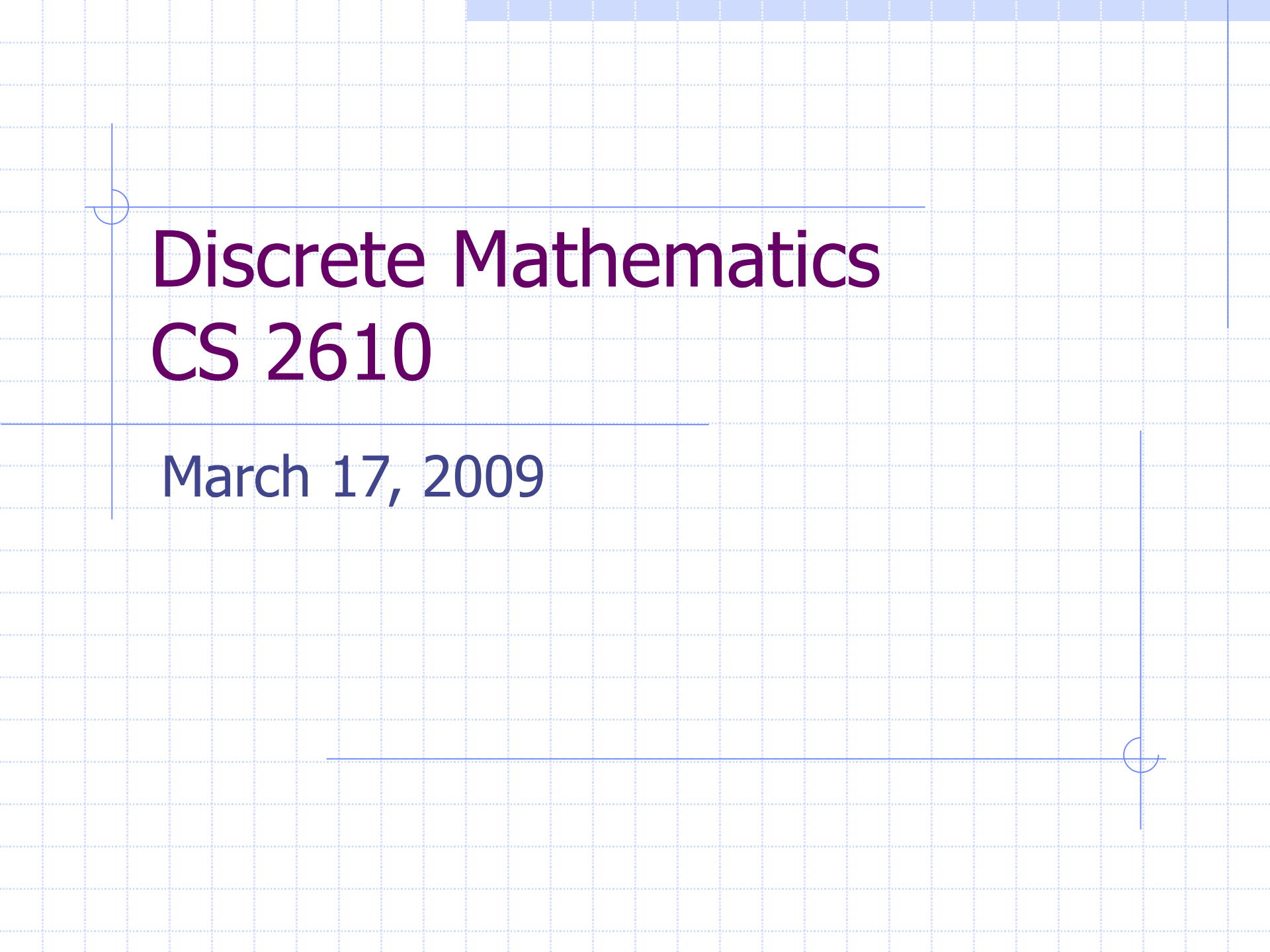




Discrete Mathematics

CS 2610

March 17, 2009



Number Theory

◆ Elementary number theory, concerned with numbers, usually integers and their properties or rational numbers

- mainly divisibility among integers
- Modular arithmetic

◆ Some Applications

- Cryptography
 - ◆ E-commerce
 - ◆ Payment systems
 - ◆ ...
- Random number generation
- Coding theory
- Hash functions (as opposed to stew functions 😊)

Number Theory - Division

Let a , b and c be integers, st $a \neq 0$, we say that "a divides b" or $a|b$ if there is an integer c where $b = a \cdot c$.

◆ a and c are said to **divide b** (or are **factors**)

$$a | b \wedge c | b$$

◆ b is a **multiple** of both a and c

Example:

$$5 | 30 \text{ and } 5 | 55 \text{ but } 5 \nmid 27$$

Number Theory - Division

Theorem 3.4.1: for all $a, b, c \in \mathbb{Z}$:

1. $a|0$
2. $(a|b \wedge a|c) \rightarrow a|(b+c)$
3. $a|b \rightarrow a|bc$ for all integers c
4. $(a|b \wedge b|c) \rightarrow a|c$

Proof: (2) $a|b$ means $b = ap$, and $a|c$ means $c = aq$

$$b + c = ap + aq = a(p + q)$$

therefore, $a|(b + c)$, or $(b + c) = ar$ where $r = p+q$

Proof: (4) $a|b$ means $b = ap$, and $b|c$ means $c = bq$

$$c = bq = apq$$

therefore, $a|c$ or $c = ar$ where $r = pq$

Division

Remember long division?

$$\begin{array}{r} 3 \\ 30 \overline{) 109} \\ \underline{90} \\ 19 \end{array}$$

$$109 = 30 \cdot 3 + 19$$

$$a = dq + r \text{ (dividend = divisor} \cdot \text{quotient} + \text{remainder)}$$

The Division Algorithm

Division Algorithm Theorem: Let a be an integer, and d be a positive integer. There are unique integers q, r with $r \in \{0, 1, 2, \dots, d-1\}$ (ie, $0 \leq r < d$) satisfying

$$a = dq + r$$

- ◆ d is the divisor

- ◆ q is the quotient

$$q = a \text{ div } d$$

- ◆ r is the remainder

$$r = a \text{ mod } d$$

Mod Operation

Let $a, b \in \mathbb{Z}$ with $b > 1$.

$$a = q \cdot b + r, \text{ where } 0 \leq r < b$$

Then $a \bmod b$ denotes the remainder r from the division "algorithm" with dividend a and divisor b

$$109 \bmod 30 = ?$$

$$\diamond 0 \leq a \bmod b \leq b - 1$$

Modular Arithmetic

◆ Let $a, b \in \mathbb{Z}, m \in \mathbb{Z}^+$

Then a is congruent (denk) to b modulo m iff $m \mid (a - b)$.

◆ Notation:

- " $a \equiv b \pmod{m}$ " reads a is congruent to b modulo m
- " $a \not\equiv b \pmod{m}$ " reads a is not congruent to b modulo m .

◆ Examples:

- $5 \equiv 25 \pmod{10}$
- $5 \not\equiv 25 \pmod{3}$

Modular Arithmetic

⊕ **Theorem 3.4.3:** Let $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$. Then
 $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$

Proof: (1) given $a \bmod m = b \bmod m$ we have

$$a = ms + r \text{ or } r = a - ms,$$

$$b = mp + r \text{ or } r = b - mp,$$

$$a - ms = b - mp$$

$$\begin{aligned} \text{which means } a - b &= ms - mp \\ &= m(s - p) \end{aligned}$$

so $m \mid (a - b)$ which means

$$a \equiv b \pmod{m}$$

Modular Arithmetic

⊕ **Theorem 3.4.3:** Let $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$. Then
 $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$

Proof: (2) given $a \equiv b \pmod{m}$ we have $m \mid (a - b)$

let $a = mq_a + r_a$ and $b = mq_b + r_b$

so, $m \mid ((mq_a + r_a) - (mq_b + r_b))$

or $m \mid m(q_a - q_b) + (r_a - r_b)$

recall $0 \leq r_a < m$ and $0 \leq r_b < m$

therefore $(r_a - r_b)$ must be 0

that is, the two remainders are the same

which is the same as saying

$a \bmod m = b \bmod m$

Modular Arithmetic

Theorem 3.4.4: Let $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$. Then:
 $a \equiv b \pmod{m}$ iff there exists a $k \in \mathbb{Z}$ st

$$a = b + km.$$

Proof: $a = b + km$ means
 $a - b = km$ which means
 $m \mid (a - b)$ which is the same as saying
 $a \equiv b \pmod{m}$
(to complete the proof, reverse the steps)

Examples:

$$27 \equiv 12 \pmod{5}$$

$$27 = 12 + 5k \quad k = 3$$

$$105 \equiv -45 \pmod{10}$$

$$105 = -45 + 10k \quad k = 15$$

Modular Arithmetic

⊕ **Theorem 3.4.5:** Let $a, b, c, d \in \mathbb{Z}$, $m \in \mathbb{Z}^+$. Then if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

1. $a + c \equiv b + d \pmod{m}$,
2. $a - c \equiv b - d \pmod{m}$,
3. $ac \equiv bd \pmod{m}$

Proof: $a = b + k_1m$ and $c = d + k_2m$

$$a + c = b + d + k_1m + k_2m$$

$$\text{or } a + c = b + d + m(k_1 + k_2)$$

which is

$$a + c \equiv b + d \pmod{m}$$

others are similar

Modular Arithmetic - examples

Hash Functions: record access scheme for finding a record very quickly based on some key value in the record. That is, there is a mapping between the key value and the memory location for the record.

Ex. $h(k) = k \bmod m$ (an onto function, why?)

k is the record's key value

m is the number of memory locations

Collisions occur since h is not one-to-one. What then? Typically, invoke a secondary hash function or some other scheme (sequential search).

Modular Arithmetic - examples

Pseudorandom numbers: generated using the linear congruential method (doğrusal eşlik metodu)

m - modulus

a - multiplier

c - increment

x_0 - seed

$$2 \leq a < m, \quad 0 \leq c < m, \quad 0 \leq x_0 < m$$

Generate the set of PRNs $\{x_n\}$ with $0 \leq x_n < m$ for all n

$$x_{n+1} = (ax_n + c) \bmod m$$

(divide by m to get PRNs between 0 and 1)

Modular Arithmetic - examples

cryptology: secret codes, encryption/decryption

Caesar encryption (positional 3-offset scheme)

For our 26 letters, assign integers 0-25

$$f(p) = (p + 3) \bmod 26$$

"PARK" maps to integers 15, 0, 17, 10 which are then encrypted into 18, 3, 20, 13 or "SDUN"

use the inverse $(p - 3) \bmod 26$ to decrypt back to "PARK"

Number Theory - Primes

A positive integer $n > 1$ is called **prime** if it is only divisible by 1 and itself (i.e., only has 1 and itself as its positive factors).

Example: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 97

A number $n \geq 2$ which isn't prime is called **composite**.
(Iff there exists an a such that $a|n$ and $1 < a < n$)

Example:

All even numbers > 2 are composite.

By convention, 1 is neither **prime** or **composite**.

Number Theory - Primes

Fundamental Theorem of Arithmetic

Every positive integer greater than 1 has a *unique* representation as the product of a non-decreasing series of one or more primes

Examples:

- ♦ $2 = 2$
- ♦ $4 = 2 \cdot 2$
- ♦ $100 = 2 \cdot 2 \cdot 5 \cdot 5$
- ♦ $200 = 2 \cdot 2 \cdot 2 \cdot 5 \cdot 5$
- ♦ $999 = 3 \cdot 3 \cdot 3 \cdot 37$

Number Theory – Primality Testing

◆ How do you check whether a positive integer n is prime?

◆ Solution:

Start testing to see if prime p divides n ($2|n$, $3|n$, $5|n$, etc). When one is found, use the dividend and begin again. Repeat.

Find prime factorization for 7007.

2, 3, 5 don't divide 7007 but 7 does (1001)

Now, 7 also divides 1001 (143)

7 doesn't divide 143 but 11 does (13) and we're done.

Number Theory - Primes

Theorem 3.5.2 : If n is composite, then it has a prime factor (divisor) that is less than or equal to $\sqrt{n}$

Proof: if n is composite, we know it has a factor a with $1 < a < n$. IOW $n = ab$ for some $b > 1$. So, either $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ (note, if $a > \sqrt{n}$ and $b > \sqrt{n}$ then $ab > n$, nope). OK, both a and b are divisors of n , and n has a positive divisor not exceeding $\sqrt{n}$. This divisor is either prime or it has a prime divisor less than itself. In either case, n has a prime divisor $\leq \sqrt{n}$.

*** An integer is prime if it is not divisible by any prime less than or equal to its square root.

Number Theory – Prime Numbers

Theorem 3.5.4: The number of primes not exceeding n is asymptotic to $n/\log n$.

i.e. $\lim_{n \rightarrow \infty} \Pi(n)/(n \log n) \rightarrow 1$

$\Pi(n)$: number of prime numbers less than or equal to n

n	$\Pi(n)$	$n/\log n$
1000	168	145
10000	1229	1086
100000	9592	8686
1000000	78498	72382
10000000	664579	620420
100000000	5761455	5428681

Number Theory – Prime Numbers

There are still plenty of things we don't know about primes:

- * no cool function gives us primes, not even

$$f(n) = n^2 - n + 41$$

- * Goldbach's conjecture (varsayım) : every even integer n where $n > 2$ is the sum of two primes
($18=13+5$; $24= 19+5$; $32=29+3$)

- * twin prime conjecture: there are infinitely many twin primes (pairs p and $p+2$, both prime)

Greatest Common Divisor

Let a, b be integers, $a \neq 0$, $b \neq 0$, not both zero.

The ***greatest common divisor*** of a and b is the biggest number d which divides both a and b .

Example: $\gcd(42, 72)$

Positive divisors of 42: 1, 2, 3, 6, 7, 14, 21

Positive divisors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 24, 36

$\gcd(42, 72) = 6$

Finding the GCD

◆ If the prime factorizations are written as

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n} \quad \text{and} \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

then the GCD is given by:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}.$$

Example:

$$\bullet a = 42 = 2 \cdot 3 \cdot 7$$

$$= 2^1 \cdot 3^1 \cdot 7^1$$

$$\bullet b = 72 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3$$

$$= 2^3 \cdot 3^2 \cdot 7^0$$

$$\bullet \gcd(42, 72)$$

$$= 2^1 \cdot 3^1 \cdot 7^0 = 2 \cdot 3 = 6$$

Least Common Multiple

The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b .

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}.$$

$$\text{Example: } \text{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^4 3^5 7^2$$

Least Common Multiple

Let a and b be positive integers. Then

$$ab = \gcd(a, b) \cdot \text{lcm}(a, b)$$

Modular Exponentiation

- ◆ Let b be base, n, m large integers, $b < m$.
- ◆ The modular exponentiation is computed as

$$b^n \bmod m$$

Fundamental in cryptography: RSA encryption

How can we compute the modular exponentiation ?

Modular Exponentiation

For large b , n and m , we can compute the modular exponentiation using the following property:

$$a \cdot b \bmod m = (a \bmod m) (b \bmod m) \bmod m$$

$$(\text{Mod}5)91 = 13 * 7 = 3 * 2 (\text{mod } 5) = 1$$

$$\text{Therefore, } b^n (\bmod m) = (b \bmod m)^n (\bmod m)$$

In fact, we can take $(\bmod m)$ after each multiplication to keep all values low.

Example

◆ Find $37^5 \pmod{5}$

$$37^5 \pmod{5} = (37 \pmod{5})^5 \pmod{5} = 2^5 \pmod{5}$$

$$\begin{aligned} 2^5 \pmod{5} &= 2*2*2*2*2 \pmod{5} = 4*2*2*2 \pmod{5} = \\ &8*2*2 \pmod{5} = 3*2*2 \pmod{5} = 6*2 \pmod{5} = \\ &1*2 \pmod{5} = 2 \pmod{5} = 2 \end{aligned}$$

Can you see a way to shorten this process?

Use results you have already calculated

$$2^5 \pmod{5} = 4*4*2 \pmod{5} = 16*2 \pmod{5} = 2$$

For large exponents this can make a **big** difference!

Cryptography

- ◆ Cryptology is the study of secret (coded) messages.
- *Cryptography* - Methods for encrypting (şifreleme) and decrypting (şifre çözme) secret messages using **secret keys**.
 - ◆ **Encryption** is the process of transforming a message to an unreadable form.
 - ◆ **Decryption** is the process of transforming an encrypted message back to its original form.
 - ◆ Both **encryption** and **decryption** require the use of some secret knowledge known as the **secret key**.
- *Cryptoanalysis* - Methods for decrypting an encrypted message without knowing the secret keys.

Cryptography - Caesar's shift cypher

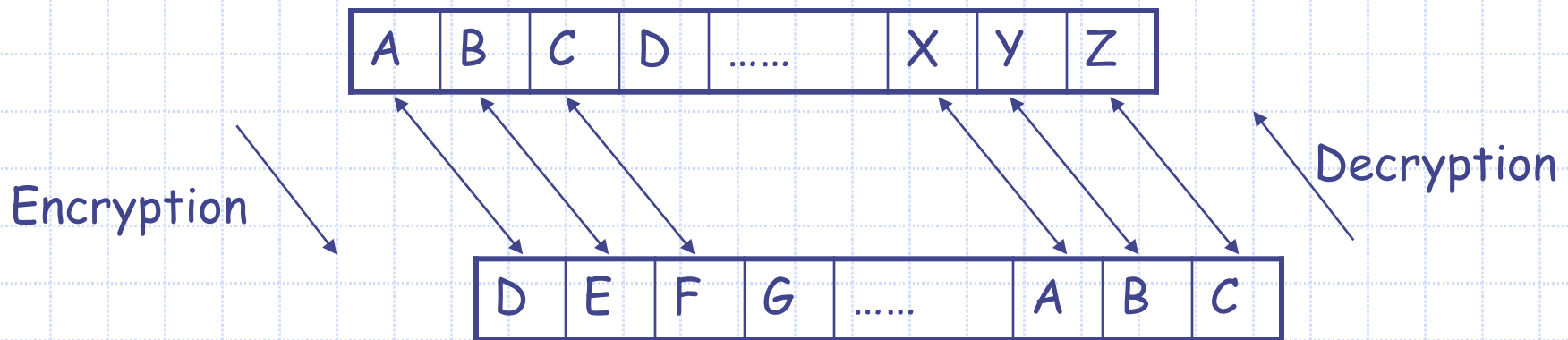
Encryption

- Shift each letter in the message three letters forward in the alphabet.

Decryption

- Shift each letter in the message three letters backward in the alphabet.

hello world → khoor zruog



Public Key Cryptography

◆ *Public key cryptosystems use two keys*

- **Public key** to encrypt the message
 - ◆ Known to everybody
- **Private Key** to decrypt the encrypted message
 - ◆ It is kept secret.
 - ◆ It is computationally infeasible to guess the Private Key

◆ RSA one of the most widely used *Public key cryptosystem*

Ronald Rivest, Adi Shamir, and Leonard Adleman

RSA Basis

- ◆ Let p and q be two large primes, and $e \in \mathbb{Z}$ such that $\gcd(e, (p-1)(q-1)) = 1$
and d (the decryption key) is an integer such that $de \equiv 1 \pmod{(p-1)(q-1)}$
- ◆ p and q are large primes, over 100 digits each.
- ◆ **Public Key**
 - $n=pq$ (the modulus)
 - e (the public exponent)
 - It is common to choose a small public exponent for the public key.
- ◆ **Private Key**
 - d (the private exponent)

RSA

◆ Encryption

- Let M be a message such that $M < n$
- Compute $C = M^e \bmod n$
 - ◆ This can be done using *Binary Modular Exponentiation*

◆ Decryption

- Compute $M = C^d \bmod pq$

Why Does RSA Work?

- ◆ The correctness of the RSA method results from the assumption that neither p nor q divides M (which will be true for most messages) and the following two theorems.

- ◆ Fermat's Little Theorem

If p is a prime and a is an integer not divisible by p then $a^{p-1} \equiv 1 \pmod{p}$.

$$P=7, a=9, 9^6 \pmod{7} = 2^6 \pmod{7} = 1$$

- ◆ The Chinese Remainder Theorem

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers. The system (aralarında asal)

$$x \equiv a_1 \pmod{m_1} \quad x \equiv a_2 \pmod{m_2} \quad \dots \quad x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m_1 m_2 \dots m_n$ - i.e., there is only one x such that $0 \leq x < m_1 m_2 \dots m_n$ that satisfies the above congruencies.

Çinli Kalan Teoremi

Çinlilerin kalan teoremi

Çinlilerin kalan teoremi, "3'e bölündüğünde 2, 5'e bölündüğünde 3, 7'ye bölündüğünde 4 kalanını veren sayıyı bulun" tipinden problemleri çözmek için kullanılan teorem. buna göre: 3'e bölündüğünde 2 kalanını veren sayılar $3k+2$ şeklindedir. (2, 5, 8, ...) 5'e bölündüğünde 3 kalanını veren sayılar $5l+3$ şeklindedir. (3, 8, 13, ...) bu durumda sayımız $15m+8$ şeklindedir. (8, 23, 38, 53, ...) 7'ye bölündüğünde 4 kalanını veren sayılar $7n+4$ şeklindedir. (4, 11, 18, ..., 46, 53, ...) bu durumda da sayımız $105p+53$ şeklindedir.

$n_1, n_2, \dots, n_k$ pozitif, çiftli aralarında asal **Tamsayı** olsun. Bu durumda, Verilen herhangi $a_1, a_2, \dots, a_k$ tamsayıları için bir x tamsayısı vardır ki sistemin eşzamanlı uygun bir çözümüdür.

$$x \equiv a_1 \pmod{n_1}$$

$$x \equiv a_2 \pmod{n_2}$$

$$\vdots$$

$$x \equiv a_k \pmod{n_k}$$

Bundan başka, Tüm Çözümler x Bu sistem uyumlu olan modulo $N = n_1 n_2 \dots n_k$.

Böylece $x \equiv y \pmod{n_i}$ tüm $1 \leq i \leq k$, ancak ve ancak $x \equiv y \pmod{N}$.

Sometimes, the simultaneous congruences can be solved even if the n_i 's are not pairwise coprime. A solution x exists if and only if:

$$a_i \equiv a_j \pmod{\gcd(n_i, n_j)} \quad \text{for all } i \text{ and } j.$$

All solutions x are then congruent modulo the least common multiple of the n_i .

Versions of the Chinese remainder theorem were also known to **Brahmagupta** (7th century), and appear in **Fibonacci's Liber Abaci** (1202).

Çinli Kalan Teoremi

İspat. $n = n_1 n_2 \cdots n_k$ olsun.

$1 \leq i \leq k$ için $\left(\frac{n}{n_i}, n_i\right) = 1$ olduğundan $\frac{n}{n_i} s_i \equiv r_i \pmod{n_i}$ denklik sistemini sağlayan s_i vardır.

$$x \equiv \sum_{i=1}^k \frac{n}{n_i} s_i \pmod{n}.$$

Aşağıdaki denklik sistemini çözünüz.

$$x \equiv 2 \pmod{7}$$

$$x \equiv 4 \pmod{9}$$

$$x \equiv 1 \pmod{10}.$$

Çözüm. $n = 7 \cdot 9 \cdot 10 = 630$, $\frac{n}{n_1} = 90$, $\frac{n}{n_2} = 70$ ve $\frac{n}{n_3} = 63$ olduğundan

$$90s_1 \equiv 2 \pmod{7} \Rightarrow 6s_1 \equiv 2 \pmod{7} \Rightarrow s_1 \equiv 6^{-1} \cdot 2 \equiv 6 \cdot 2 \equiv 5 \pmod{7}$$

$$70s_2 \equiv 4 \pmod{9} \Rightarrow 7s_2 \equiv 4 \pmod{9} \Rightarrow s_2 \equiv 7^{-1} \cdot 4 \equiv 4 \cdot 4 \equiv 7 \pmod{9}$$

$$63s_3 \equiv 1 \pmod{10} \Rightarrow 3s_3 \equiv 1 \pmod{10} \Rightarrow s_3 \equiv 3^{-1} \cdot 1 \equiv 7 \pmod{10}$$

bulunur. Buradan $x \equiv \sum_{i=1}^3 \frac{n}{n_i} s_i = 90 \cdot 5 + 70 \cdot 7 + 63 \cdot 7 = 1381$ olur. Çözüm mod 630 da tek olduğundan $x = 1381 \equiv 121 \pmod{630}$ elde edilir.

Why Does RSA Work?

- ◆ Since $de \equiv 1 \pmod{(p-1)(q-1)}$, we can conclude that $de = 1 + k(p-1)(q-1)$.
- ◆ Therefore $C^d \equiv (M^e)^d = M^{de} = M^{1+k(p-1)(q-1)} \pmod{n}$.
- ◆ Assuming $\gcd(M, p) = \gcd(M, q) = 1$, we can conclude (by Fermat's Little Theorem) that $M, p-1$ 'e tam bölünmüyor (1 kalıyor)
 - $C^d \equiv M \cdot (M^{p-1})^{k(q-1)} \equiv M \cdot 1 \equiv M \pmod{p}$
 - $C^d \equiv M \cdot (M^{q-1})^{k(p-1)} \equiv M \cdot 1 \equiv M \pmod{q}$
- ◆ By the Chinese Remainder Theorem, we can conclude that
 - $C^d \equiv M \pmod{pq}$
 - ◆ Recall that $n = pq$

RSA Example

- ◆ Let $p = 61$ and $q = 53$
 - Then $n = pq = 3233$
- ◆ Let $e = 17$ and $d = 2753$
 - Note $17 * 2753 = 46801 = 1 + 15 * 60 * 52$
- ◆ Public keys: e, n
- ◆ Private key: d
- ◆ Encrypt 123
 - $123^{17} \pmod{3233} = 855$
- ◆ Decrypt 855
 - $855^{2753} \pmod{3233} = 123$, tek bir d var bunu sağlayan
- ◆ We need clever exponentiation techniques!

Breaking RSA

How to break the system

1. An attacker discovers the numbers p and q
 - Find the prime factorization of n
 - Computationally difficult when p and q are chosen properly.
 - The modulus n must be at least 2048 bits long
 - ◆ On May 10, 2005, **RSA-200**, a 200-digit number module was factored into two 100-digit primes by researchers in Germany
 - The effort started during Christmas 2003 using several computers in parallel.
 - Equivalent of 55 years on a single 2.2 GHz Opteron CPU

RSA – In practice

◆ How to break the system

2. Find e -th roots mod n .

- The encrypted message C is obtained as

- $C = M^e \bmod n$

- No general methods are currently known to find the e -th roots mod n , except for special cases.

Review of Secret Key (Symmetric) Cryptography

- ◆ Confidentiality
 - stream ciphers (uses PRNG)
 - block ciphers with encryption modes
- ◆ Integrity
 - Cryptographic hash functions
 - Message authentication code (keyed hash functions)
- ◆ Limitation: sender and receiver must share the same key
 - Needs secure channel for key distribution
 - Impossible for two parties having no prior relationship
 - Needs many keys for n parties to communicate

Public Key Encryption Overview

- ◆ Each party has a PAIR (K, K^{-1}) of keys:
 - K is the **public** key, and used for encryption
 - K^{-1} is the **private** key, and used for decryption
 - Satisfies $D_{K^{-1}}[E_K[M]] = M$
- ◆ Knowing the public-key K , it is computationally infeasible to compute the private key K^{-1}
 - **How to check (K, K^{-1}) is a pair?**
 - Offers only computational security. PK Encryption impossible when $P=NP$, as deriving K^{-1} from K is in NP.
- ◆ The public-key K may be made publicly available, e.g., in a publicly available directory
 - Many can encrypt, only one can decrypt
- ◆ Public-key systems aka **asymmetric** crypto systems

Public Key Cryptography Early History

- ◆ The **concept** is proposed in Diffie and Hellman (1976) "New Directions in Cryptography"
 - public-key encryption schemes
 - public key distribution systems
 - ◆ Diffie-Hellman key agreement protocol
 - digital signature
- ◆ Public-key encryption was proposed in 1970 by James Ellis
 - in a classified paper made public in 1997 by the British Governmental Communications Headquarters
- ◆ Concept of digital signature is still originally due to Diffie & Hellman

Public-Key Encryption Algorithms

- ◆ Almost all public-key encryption algorithms use either number theory and modular arithmetic, or elliptic curves
- ◆ RSA
 - based on the hardness of factoring large numbers
- ◆ El Gamal
 - Based on the hardness of solving discrete logarithm
 - Basic idea: public key g^x , private key x , to encrypt: $[g^y, g^{xy} M]$.

RSA Algorithm

- ◆ Invented in **1978** by Ron Rivest, Adi Shamir and Leonard Adleman
 - Published as R L Rivest, A Shamir, L Adleman, "On Digital Signatures and Public Key Cryptosystems", Communications of the ACM, vol 21 no 2, pp120-126, Feb 1978
- ◆ Security relies on the difficulty of factoring large composite numbers
- ◆ Essentially the same algorithm was discovered in 1973 by Clifford Cocks, who works for the British intelligence

RSA Public Key Crypto System

Key generation:

1. Select 2 large prime numbers of about the same size, p and q
Typically each p, q has between 512 and 2048 bits
2. Compute $n = pq$, and $\Phi(n) = (q-1)(p-1)$
3. Select e , $1 < e < \Phi(n)$, s.t. $\gcd(e, \Phi(n)) = 1$
Typically $e=3$ or $e=65537$
4. Compute d , $1 < d < \Phi(n)$ s.t. $ed \equiv 1 \pmod{\Phi(n)}$
Knowing $\Phi(n)$, d easy to compute.

Public key: (e, n)

Private key: d

RSA Description (cont.)

Encryption

Given a message M , $0 < M < n$ $M \in \mathbb{Z}_n - \{0\}$
use public key (e, n)
compute $C = M^e \bmod n$ $C \in \mathbb{Z}_n - \{0\}$

Decryption

Given a ciphertext C , use private key (d)
Compute $C^d \bmod n = (M^e \bmod n)^d \bmod n = M^{ed} \bmod n =$
 M

Plaintext:
M

$$C = M^e \bmod (n=pq)$$

Ciphertext: C

$$C^d \bmod n$$

From n , difficult to figure out p, q

From (n, e) , difficult to figure d .

From (n, e) and C , difficult to figure out M s.t. $C = M^e$

RSA Example

- ◆ $p = 11, q = 7, n = 77, \Phi(n) = 60$
- ◆ $d = 13, e = 37$ ($ed = 481; ed \bmod 60 = 1$)
- ◆ Let $M = 15$. Then $C \equiv M^e \bmod n$
 - $C \equiv 15^{37} \bmod 77 = 71$
- ◆ $M \equiv C^d \bmod n$
 - $M \equiv 71^{13} \bmod 77 = 15$